

DriveBuddyAI – Pav Bhaji Classification

Text-Based Classification Using Instagram Post Metadata

Machine Learning Challenge Submission

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1. Project Summary

The objective of this project is to classify whether an Instagram post is related to Pav Bhaji using the textual metadata provided with the challenge. Following the challenge requirement, the solution does **not** use image pixels or a CNN. Instead, caption and tag information is transformed into numerical text features and used to train classical machine-learning classifiers.

The final pipeline covers data-label matching, text preprocessing, TF-IDF feature extraction, chi-square feature selection, model comparison, class-balanced training and evaluation on an unseen stratified test set.

2. Dataset and Label Preparation

The supplied JSON metadata contained 1,500 Instagram records. After matching the metadata with the labels available from the supplied image folders, 452 records were available for supervised learning.

Dataset characteristic	Value
Total metadata records	1,500
Matched labelled records	452
Non-Pav Bhaji	269 (59.5%)
Pav Bhaji	183 (40.5%)
Train / test split	80% / 20%
Training records	361
Test records	91

3. Methodology

Step 1 – Text construction: Caption and tags were combined into a single text representation for each post.

Step 2 – TF-IDF: The text was represented using TF-IDF with unigrams and bigrams. This produced 3,495 text features before selection.

Step 3 – Feature selection: Chi-square feature selection was applied to retain the 1,000 most useful features for classification.

Step 4 – Model comparison: Logistic Regression, Linear SVM and Naive Bayes were evaluated, followed by class-balanced Logistic Regression and Linear SVM to address the class distribution.

Step 5 – Evaluation: The final model was evaluated on the held-out test set using accuracy, precision, recall, F1-score, ROC-AUC and a confusion matrix.

4. Model Comparison

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.593	0.500	0.216	0.302
Linear SVM	0.560	0.452	0.378	0.412
Naive Bayes	0.582	0.481	0.351	0.406
Balanced Logistic Regression	0.626	0.527	0.784	0.630
Balanced Linear SVM	0.604	0.511	0.649	0.571

The class-balanced Logistic Regression model was used as the final model because it provided higher positive-class recall and F1-score than the other evaluated models in the held-out test set.

5. Final Evaluation

Metric	Test-set result
Accuracy	62.6%
Precision	52.7%
Recall	78.4%
F1 Score	63.0%
ROC-AUC	65.4%

6. Confusion Matrix

Actual / Predicted	Non-Pav Bhaji	Pav Bhaji
Non-Pav Bhaji	28	26
Pav Bhaji	8	29

On the 91-record test set, the final model correctly classified 29 Pav Bhaji posts and missed 8. It also correctly classified 28 non-Pav Bhaji posts while producing 26 false positives. This pattern is consistent with the relatively high recall and more moderate precision of the final balanced model.

7. Feature Interpretation

The trained Logistic Regression model assigns coefficients to TF-IDF features. Some of the observed positive-side terms included **pav**, **bhukkad**, **breakfast**, **butter**, **vegetables**, **buttery** and **mumbaifood**, while terms such as **vadapav**, **kolkata**, **delhigram** and **kolkatadiaries** appeared among the negative-side features.

These coefficients indicate which words were associated with the model's predictions in this dataset; they should not be interpreted as causal relationships or universal indicators of Pav Bhaji.

8. Practical Output

The notebook includes a prediction function that accepts a caption and tags as input and returns either **Pav Bhaji** or **Not Pav Bhaji**. This provides a simple interface for testing new Instagram-style text inputs after the model has been trained.

9. Limitations and Future Scope

The supervised dataset contains 452 matched labelled records, so the model should be treated as a baseline text-classification solution rather than a production-ready classifier. Social-media text is also noisy, and vocabulary

can vary across users and locations.

Possible future work includes collecting more labelled metadata, improving text normalisation, testing additional linear classifiers, tuning hyperparameters with cross-validation, and evaluating robustness on a separate external dataset.

10. Conclusion

This project implements the required text-based Pav Bhaji classification workflow using the metadata supplied with the DriveBuddyAI challenge. The final class-balanced Logistic Regression model achieved **62.6% accuracy, 78.4% recall, 63.0% F1-score and 65.4% ROC-AUC** on the held-out test set.

The main outcome is a complete and reproducible machine-learning pipeline from metadata preparation to model evaluation, while remaining aligned with the challenge requirement of using textual information rather than image-based classification.

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